

Quantitative Research

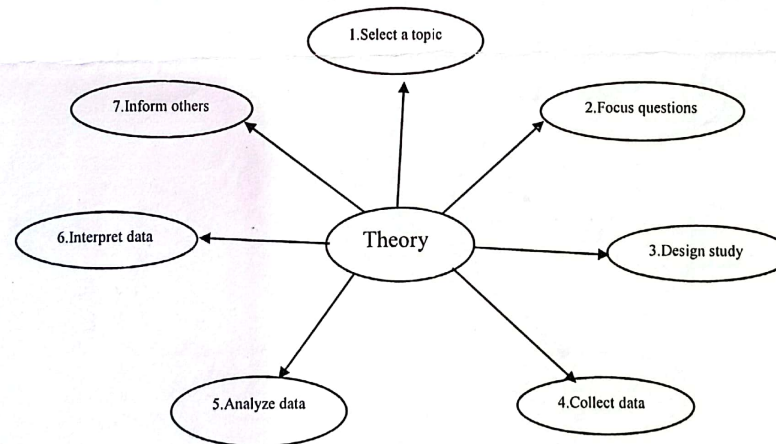
Definition:

"An inquiry into a social or human problem, based on testing a theory composed of variables, measured with numbers, and analyzed with statistical procedures, in order to determine whether the predictive generalization of the theory hold true." **Cresswell (2014)**

Following are the characteristics of Quantitative research:

- Tools: Surveys, structured interviews & observations, and reviews of records or documents.
- Primarily deductive (general to specific) process used to test pre-specified concepts, constructs and hypothesis that make up a theory.
- More objective: provide observed effects (interpreted by researchers) of a program on a problem or condition.
- Numbered-based.
- Less in depth but more breadth of information across a large number of cases.
- Fixed response options.
- Statistical tests are used for analysis.
- Can be valid and reliable: largely depends on the measurement device or instrument used.
- Time expenditure heavier on the planning phase and lighter on the analysis phase.
- More generalizable.
- In Quantitative research, researchers use deductive approach (General to specific).

STEPS IN THE QUANTITATIVE RESEARCH PROCESS



Types of quantitative research

There are four main types of quantitative research.

1. Descriptive.
2. Correlational comparative research
3. Causal-Comparative/Quasi-Experimental.
4. And Experimental Research.
5. Quasi-Experimental research

Qualitative research

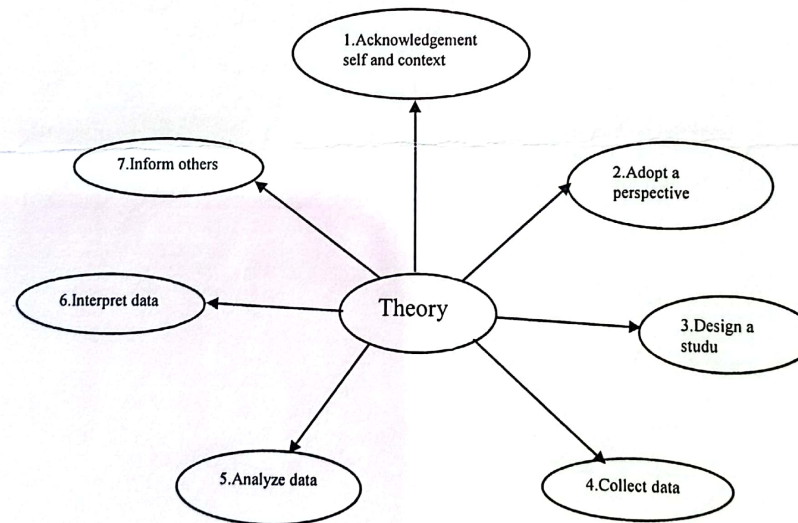
Definition:

Qualitative research is rooted in the social sciences and is concerned with people and their social realities with how the social world is understood, experienced, interpreted, and constituted with individual and collective meaning. Interpretations behaviors and social processes. **Bryman (2004)**

Following are the characteristics of quantitative research:

- Tools: Focus groups, in-depth interviews, and reviews of documents for types of themes and observation.
- Primarily inductive (specific to general) process used to formulate theory or hypothesis.
- More subjective: describe a problem or condition from the point of those who experiencing it.
- Text-based.
- More in depth information on a few cases.
- Unstructured or semi-structured response options.
- No statistical tests.
- Can be valid and reliable: largely depends on skill and rigor of the researcher.
- Time expenditure lighter on the planning phase and heavier during the analysis phase.
- Less generalizable.

STEPS IN QUALITATIVE RESEARCH PROCESS



Types of qualitative research

There are six main types of qualitative research:

1. Phenomenological.
2. Ethnographic.
3. Grounded theory.
4. Historical.
5. Case study.
6. And action research.

The Nature of Educational Research

1.1 Definition of educational research

✓ Research

Research is the systematic investigation and study of materials and sources in order to establish facts and reach new conclusions.

Research is the foundation of knowledge for the purpose of knowledge and an important source for providing guidelines or norms for solving different social, business, or governmental problems.

Research is "creative and systematic work undertaken to increase the stock of knowledge". It involves the collection, organization and analysis of evidence to increase understanding of a topic, characterized by a particular attentiveness to controlling sources of bias and error.

SIR ✓ Educational research

Educational research refers to the systematic collection and analysis of data related to the field of education. Research may involve a variety of methods and various aspects of education including student learning, interaction, teaching methods, teacher training, and classroom dynamics.

Here are some definitions of educational research according to different authors

Crawford:

"Educational research is a systematic and refined technique of thinking, using special tools in order to obtain a mere adequate solution of a problem."

|| C.V. Good: ||

"Educational research is the study and investigation in the field of education."

|| Munroe: ||

Qudsia Zafar

- Related to field of education.
- Involve a variety of methods
- Various aspects of Educat.
- 1) Student learning
- 2) Teaching Methods
- 3) Teacher Training
- 4) Classroom dynamics

Research in education

"The final purpose of educational research is to ascertain principles and develop procedures for use in the field of education."

} Ascertain Principles } To use
} Develop Procedures } in field
of education

Mulay:

"Any systematic study designed to promote the development of education as a science can be considered educational research."

J. W. Best:

"Educational research is that activity which is directed towards development of a science of behaviour in educational situations. The ultimate aim of such a science is to provide knowledge that will permit the educator to achieve his goals by the most effective methods."

W. M. Traverse:

"Educational research is that activity which is directed towards the development of science of behaviour in educational situations."

Whitney:

"Educational Research aims at finding out solution of educational problems by using scientific philosophical method."

Educators need to be consumers

and producers of education

✓ Characteristics of educational research

- ✓ It sets out to solve a specific problem
- ✓ Educational research adopts primary and secondary research methods in its data collection process. This means that in educational research, the investigator relies on first-hand sources of information and secondary data to arrive at a suitable conclusion.
 - Educational research relies on empirical evidence. This results from its largely scientific approach.
- ✓ Educational research is objective and accurate because it measures verifiable information.

- ✓ In educational research, the researcher adopts specific methodologies, detailed procedures, and analysis to arrive at the most objective responses
 - Educational research findings are useful in the development of principles and theories that provide better insights into pressing issues.
- ✓ This research approach combines structured, semi-structured, and unstructured questions to gather verifiable data from respondents.
 - Many educational research findings are documented for peer review before their presentation.
 - Educational research is interdisciplinary in nature because it draws from different fields and studies complex factual relations.
- ✓ Educational research is planned, systematic and highly methodical in its approach.
- ✓ Educational research is a continuous inquiry in terms of knowledge.
 - Educational research is interdisciplinary in nature because it is drawn from different fields and studies

✓ 1.2 Scope and Importance

With the help of Research we introduce new Scope policies and programs.

Scope is the extent of the area or subject matter that something deals with or to which it is relevant.

The scope of a study explains the extent to which the research area will be explored in the work and specifies the parameters within the study will be operating. Basically, this means that you will have to define what the study is going to cover and what it is focusing on.

S/Q Scope of educational research

There is a wide scope of educational research including the area of educational philosophy, psychology, sociology, teacher education, curriculum development, measurement and evaluation, management and administration, comparative education, inclusive education, primary education, secondary education, higher education, educational technology, educational policies and governance, student and learning environment, etc.

→ Government / Ministry Work.

Priority Areas in Research

- Education of Backward persons.

Research in education

- Free and compulsory education
- Dropout rate
- Equal opportunities
- Population Education
- Education and handicapped
- Education of Talented persons
- Education for economic growth
- Co-ordination of all agencies in Education
- Organisation of school meals
- Follow up studies of old students
- Homework and study habits
- Student's welfare schemes, etc.

Advantages of Educational Research

Educational research helps teachers to understand children in their classroom in order to improve teaching learning process. Educational research has role of knowledge, value, beliefs. Educational research helps to bring change in dynamic structure of the education. Educational research helps analyse and solve the problems faced by educational administration. It has impact on economic progress of education.

There are tons of research papers conducted in the previous two years of not just highlighting the issues but also provide valuable solutions. there are three major scopes of educational research:

- ✓ 1. For educational faculty in schools and colleges
- ✓ 2. For solving the modern issues faced by the educational system
- ✓ 3. For bringing advancement in already existing structure for betterment

Importance of educational Research

- ✓ Educational research adds to our knowledge.
- ✓ Research improves practice.
- ✓ Educational research investigates the behaviour of the students, teachers, parents and administration that interact with educational institutions.

Qudsia Zafar

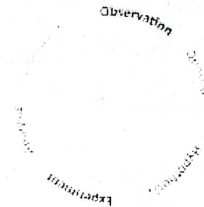
- ✓ Educational research plays a crucial role in knowledge advancement across different fields of study.
- ✓ Educational research improves teaching and learning methods by empowering you with data to help you teach and lead more strategically and effectively.
- ✓ It provides answers to practical educational challenges using scientific methods.
- ✓ Findings from educational research; especially applied research, are instrumental in policy reformulation.
- ✓ For the researcher and other parties involved in this research approach, educational research improves learning, knowledge, skills, and understanding.
- ✓ Educational research improves teaching and learning methods by empowering you with data to help you teach and lead more strategically and effectively.
- ✓ Educational research helps students apply their knowledge to practical situations.
- ✓ During the last two decades, great changes have taken place as a result of the rapid scientific and technological developments.
- ✓ Educational research helps students apply their knowledge to practical situations.
- ✓ Educational researches lead man towards progress and improve educational practice.
- ✓ Educational research help design curriculum.
- ✓ Educational research contributes to educational development.
- ✓ Educational researchers also provide valuable insight into how individual teachers do their jobs effectively to maximize their effectiveness.
- ✓ Educational research improves the quality of education.
- ✓ Researchers also look at broader issues in education. For example, some researchers study how gender affects learning styles or how students' home lives influence their academic success.

✓ Educational research helps people to decide which opinions are correct by providing evidences.

✓ 1.3 Scientific Method

Scientific method

The scientific method is an empirical method for acquiring knowledge that has characterized the development of science. It involves careful observation, applying rigorous skepticism about what is observed, given that cognitive assumptions can distort how one interprets the observation.



Scientific method in educational research

The scientific method is a step-by-step procedure for studying things; it is a systematic approach to learning about phenomena in education and other disciplines. In education, as in other disciplines, the scientific method is used to test, evaluate, and refine theories.

The scientific method is the process of objectively establishing facts through testing and experimentation. The following steps listed for the scientific method vary from text to text but usually include:

- ✓ a) define the problem,
- ✓ b) gather background information,
- ✓ c) form a hypothesis,
- ✓ d) make observations,
- ✓ e) test the hypothesis,
- ✓ f) draw conclusions.

Research
The use of scientific method teaches the student to look at all the evidence before making a statement of fact.

Characteristics of scientific method

Objectivity
Reliability
Validity
Rigor
Testability and generality
Precision
Falsifiability

- Characteristics of scientific method

Explanation

When one applies the Scientific Method to testing theories (or hypotheses), five general steps are followed:

1. Recognition or identification of the problem (something that needs to be understood better or tested);
2. formulation of hypothesis (expectations about what ought or will happen—usually hypotheses are formulated from theories, so if there is no theory, it is often difficult to determine proper hypotheses or expectations);
3. collect data that will enable one to test the hypotheses formulated above (data are the information one uses to make judgments about the problems and hypotheses);
4. analyze the data (one analyzes data in some systematic fashion in order to better understand whether the data do or do not support the hypotheses).

5. statement of findings or conclusions regarding the hypotheses are presented. Implications about the theory, based upon the tenability of the hypotheses, are drawn— i.e., do the findings support the hypotheses which in turn support the theory?

 Example

Application of the Scientific Method to the
Analysis High School Dropouts and Dropout Prevention Programs

Suppose a local high school receives a \$1,000,000 grant to develop a dropout prevention program for Bulloch county. How should the money be spent; what factors should be targeted to help prevent student dropouts?

Using a systematic approach to gaining information to answer this question is the first step, so we will use the SM to answer this question.

1. problem: Students leave high school before graduation, and society has determined that this is a problem.
2. hypothesis or expectations: Some prior research suggests a number of reasons students leave school before graduation: poor grades, lack of interest, peer pressure, financial reasons, etc. We build our expectations of the causes we have noted from our experience dealing with high school students, and from the research literature. We will want to identify factors causing students to drop out so we can better serve them.
3. data collection: Several approaches exist. We could interview students, both those who remained in school and those who left, to ask why they stayed or dropped out, then compare responses in hopes differences will emerge. Differences could help us identify key reasons for leaving school. Another approach is to assess various educational, psychological, and financial reasons via questionnaires and scales, and perform statistical analyses from a large sample to learn which of these factors best predicts school dropouts.
4. analysis of the data: For the interview data we code responses then assess common themes and differences among participants. We may find, for example, that 70% of all dropouts have grades that are very low, and that 50% of all dropouts were retained at least once in a grade. So we may see that two primary problems for these students is academic performance which leads to grade retention. For the questionnaire/scale

Research in education

data, we could build a statistical similar to the school persistence model shown above and statistically test which variables are most important for prediction dropouts. Results from these analyses may help us identify factors that can be addressed to help prevent large numbers of school dropouts.

5. conclusions: Given our findings, we may conclude that money for the dropout prevention program would best be spent on elementary school children since that is where poor school performance is first noticed and since that is where students are most likely to be retained a grade.

Steps involved in educational research

Following are the steps involved in educational research

1. identification and defining of the research problem
2. review of the related literature
3. formulation of hypothesis (assumption)
4. preparation of the research methodology / design
5. collection of the relevant data and information
6. organizing, scoring, analyzing and interpretation of data
7. draw valid conclusion and reporting the result

1.4 Research Problem and Topic

Research problem

A research problem is a specific issue or gap in existing knowledge that you aim to address in your research. You may choose to look for practical problems aimed at contributing to change, or theoretical problems aimed at expanding knowledge.

A research problem is exactly the problem that someone would like to research. A problem can be anything that a person finds unsatisfactory or unsettling, a difficulty of some sort, a state of *affairs that might be changed*, anything that is not working as well as it might.

Research in education

Problem involves areas of concern to researchers, conditions they want to improve, difficulty they want to eliminate, questions for which they seek answers.

The good examples for research problems can be found in our education setting very easily as such as

- The problems caused by at-risk students in classrooms
- Increase in violence on college and universities campuses
- Lack of parental involvement in educational process

To locate a research problem a researcher can ask various questions from themselves. They can ask themselves

- What are the issues problems are the controversies that they want to address?
- What controversies lead to the need of the study?
- What is the major concern behind conducting the study?
- What is the significance of the study?

In research article research problem can be located in the introduction section of the study. Research problem can help the policy makers when they make decisions, help teachers and school officials to solve practical problems and provide researcher with deeper understanding of educational issues.

Research topic

A research topic is the subject of a research project or study

For example: A dissertation or thesis. A research topic typically takes the form of a problem to be solved, or a question to be answered. A good research topic should be specific enough to allow for focused research and analysis.

A research topic is a subject or issue that a researcher is interested in when conducting research. A well-defined research topic is the starting point of every successful research project. Choosing a topic is an on-going process by which researchers explore, define, and refine their ideas.

Topic is a section of research problem the statement of the problem section includes five steps

- The topic (Hook Word)
- Research problem
- Justification of the importance of research problem
- The deficiencies in our existing knowledge about the problem
- How the audience will get benefit from the study of the problem

How to write a research question

- Choose your topic.
- Do some preliminary reading about the current state of the field.
- Narrow your focus to a specific niche.
- Identify the research problem that you will address.

Research topic should be general but research problem should be specific. In short, a research topic is something to be understood; a research problem is something that needs to be investigated.

1.5 Characteristic of Research Problem

Following are the characteristics of research problem

- ✓ A research problem statement is a clear, concise, and specific statement that describes the issue or problem that the research project addresses.
- ✓ It should be written in a way that is easily understandable to both experts and non-experts in the field.
- ✓ Variables in the research problem must be clear.
- ✓ Research problem must be limited in scope but specific.
- ✓ It must have some goals.
- ✓ Good researcher must be reachable.
- ✓ Research problem should be stated clearly and concisely

Research in education

- ✓ Research problem should generate research question.
- ✓ Research problem should be related to ones field of study.
- ✓ Research problem should be new that is not answered yet sufficiently by other researchers.
- ✓ The research problem must be based on the available research literature.
- ✓ Good research problem must be based on grounded on solid theory or conceptual framework
- ✓ Research problem must be controlled. Control in research refers to a methodology that is to be applied while conducting research
- ✓ Good research acknowledges previous research on the topic.
- ✓ Good research uses relevant, empirical data and proper data analysis methods.
- ✓ It has a base in the research literature.
- ✓ It has potential significance/ importance.
- ✓ It is do-able within the time frame, budget. *based on reality.*
- ✓ The research project is based on actual facts and evidence (non-hypothetical)
- ✓ It must address the gap in knowledge.
- ✓ It must be significant to the extent that it contributes positively to the research.
- ✓ It must help in further research
- ✓ The collected empirical data confirm the clarity and understand-ability of the research problem.
- ✓ It must be in the researcher's interest should and suit his/her time, practical knowledge, research skills, and resources
- ✓ The problem-solving approach must be ethical
- ✓ Customary research methods can be applied
Different methods.

Explanation

1 The Problem Can Be Stated Clearly and Concisely

Unless the problem can be stated clearly and concisely it is probably a poor problem or a non-problem. The best way to test the problem statement is to write it into a concise sentence or paragraph and to share it with others. If the problem cannot be stated in a clear paragraph it has difficulties and will not endure as a suitable problem. Of course, it is not easy to express complex issues in simplistic terms and it may take many weeks and countless drafts before the statement is satisfactory. Good critics are essential.

2 The Problem Generates Research Questions

The problem should generate a number of more specific research questions. Formulating these questions can be a challenge, particularly specifying them at the right level of abstraction.

3 It Is Grounded in Theory

Good problems have theoretical and/or conceptual frameworks for their analysis. They relate the specifics of what is being investigated to a more general background of theory which helps interpret the results and link it to the field.

4 It Relates to One or More Academic Fields of Study

Good problems relate to academic fields which have adherents and boundaries. They typically have journals to which adherents relate. Research problems which do not have clear links to one or two such fields of study are generally in trouble.

5 It Has a Base in the Research Literature

Related to the former points, a well-stated problem will relate to a research literature. With some problems, it might at first be difficult to establish the connections and literature base, but there should be a base somewhere.

6 It Has Potential Significance/importance

This is the important 'so what' question: Who cares once you solve the problem? Assume that you have solved the problem and answered the questions and then ask yourself if you are any further ahead. At the very least, the problem must have importance to the researcher, but ideally it should also be of consequence to others.

7 It Is Do-able Within the Time Frame, Budget

There are logistic factors in terms of your ability actually to carry out the research. There is no point pursuing a problem which is not feasible to research.

8 Sufficient Data Are Available or Can Be Obtained

Sometimes, there are insufficient data to address the problem. Some data banks have been developed and contain many opportunities for exploration of new questions and issues.

The researching professionalResearching professional

Researching professional refers to a person who devotes his time to gather and analyse data and provide solution to problems, a person who is skilled in conducting research and collecting information and adding advancement to knowledge. That person can be in any field of study.

2.1 Qualities of a researcherQuality

Quality is the standard of something as measured against other things of a similar kind; the degree of excellence of something.

Researcher

Researcher is a person who carries out academic or scientific research.

Researcher is a person whose job involves discovering or verifying information for use in a book, programme, etc.

Qualities of a researcher**1. Friendly with Respondents:**

A good researcher must have the quality to become friendly with respondents. It should have to talk to them in the same language in which the responding are answering and make happy made.

2. Least Discouragement:-

If the people are not co-operate to give correct data, the researcher should not be discouraged and face the difficulties, it would be called a good researcher.

3. Free From Prejudice:-

A researcher would be good if he has no prejudice or bias study about a problematic situation but he is capable of providing clear information's.

4. Capacity of Depth Information:-

A researcher should have the capacity to collect more and more information in little time.

5. Accuracy:

A researcher would be said to be good if he is accurate in his views. His ideas must be accurate one.

6. Truthful:

A researcher must have to be truthful. Its idea would be free from false reports and saying information.

7. Keen Observer:

It is the quality of a good researcher that he may have the ideas of keen and deep observation.

8. Careful in Listening: → *understanding*

A researcher would be more careful in listening. He would have the quality of listening very low information's even whispering.

9. Low Dependency on Common Sense:

A researcher should be called good if he has low dependency on common sense but keep in observation all the events and happenings.

10. Least time Consumer:

Good researcher must have the capacity of least time consuming. It will have to do more work in a little time because of the shortage of time.

11. Economical:

Good researcher must have control over his economic resources. He has to keep his finances within limits and spend carefully.

12. Low Care of Disapprovals of Society:

A good researcher have no care of the approvals or disapprovals but doing his work with zeal and patience to it.

13. Expert in Subject:

A researcher would be a good one if he has full command over his subject. He makes the use of his theoretical study in field work easily.

14. Free From Hasty Statements:

It is not expected from a good researcher to make his study hasty and invalid with wrong statements. Its study must be based on reality & validity.

15. Good in Conversation:

The conversation of a good researcher should be sympathetic and not boring. He must have the skill and art to be liked by the people.

16. Having Clear Terminology:

Research in education

A good researcher's terminology would be clear. It would be free from outwards to become difficult for the respondents to answer.

17. Trained in Research Tools:

Research is impossible without its techniques and tools. So, it should be better for a researcher to know about the use of these tools.

18. More Analytical:

A researcher would be different from other people of the society. On the basis of this quality he may observe the situation very well. Then he should be able to solve the problems easily.

19. Equality and Justice:

A good researcher should believe on equality and justice. As equal to all type of people he may collect better information's from the respondents.

20. Awareness of the possible drawbacks:

Awareness of the possible drawbacks and shortcomings of research is very essential on the part of a good researcher. By knowing it before, the researcher may try to minimize such problems, although it is well high impossible to claim complete perfection of a research work.

21. Curiosity:

A researcher should possess a strong sense of curiosity and have a genuine interest in exploring new ideas and knowledge.

22. Critical thinking:

Researchers need to be able to analyze information and data critically, evaluating its reliability and validity. They should have the ability to make logical connections, identify patterns, and draw sound conclusions.

23. Perseverance:

Research often involves extensive time and effort, as well as facing obstacles and setbacks. Researchers need to be persistent and willing to overcome challenges in order to achieve results.

24. Attention to detail:

Researchers must have a keen eye for detail and precision, as even the smallest overlooked detail can impact the validity and accuracy of research findings.

25. Open-mindedness:

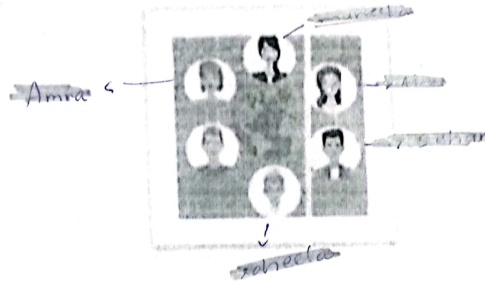
Being open minded allows researchers to consider different perspectives and approaches, encouraging innovative and creative thinking.

26. Ethical conduct:

Researchers are responsible for maintaining high ethical standards in all aspects of their work, including data collection, analysis, and reporting. This involves ensuring the privacy, confidentiality, and welfare of participants.

27. Problem-solving:

Research often involves addressing various research questions and finding solutions to problems. Researchers need to be skilled in problem-solving techniques and be able to think critically and creatively to solve research-related challenges.



2.2 Teacher as a researcher

Teacher

A teacher, formally an educator, is a person who helps students to acquire knowledge, competence, or virtue, via the practice of teaching. Informally the role of teacher may be taken on by anyone.

Researcher

Researcher is a person who carries out academic or scientific research.

Researcher is a person whose job involves discovering or verifying information for use in a book, programme, etc.

Teacher as a researcher

The concept of teacher-as-researcher is included in recent literature on educational reform, which encourages teachers to be collaborators in revising curriculum, improving their work environment, professionalizing teaching, and developing policy. One of the many benefits of teachers researching their own practices is that the teacher has a sense of ownership and control of the research because what is being researched occurs in that teacher's own classroom.

Research in education

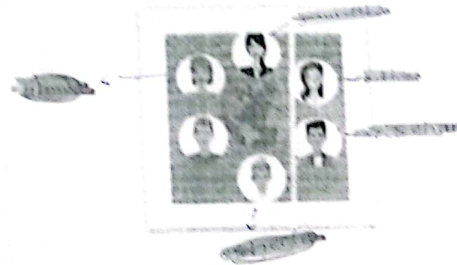
Being open-minded allows researchers to consider different perspectives and approaches, encouraging innovative and creative thinking.

26. Ethical conduct:

Researchers are responsible for maintaining high ethical standards in all aspects of their work, including data collection, analysis, and reporting. This involves ensuring the privacy, confidentiality, and welfare of participants.

27. Problem-solving:

Research often involves addressing various research questions and finding solutions to problems. Researchers need to be skilled in problem-solving techniques and be able to think critically and creatively to solve research-related challenges.



2.2 Teacher as a researcher

Teacher

A teacher, formally an educator, is a person who helps students to acquire knowledge, competence, or virtue, via the practice of teaching. Informally the role of teacher may be taken on by anyone.

Researcher

Researcher is a person who carries out academic or scientific research.

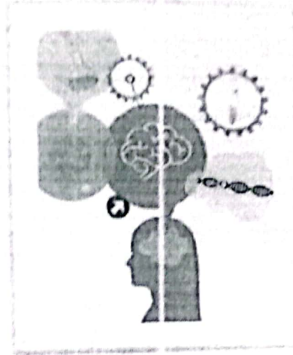
Researcher is a person whose job involves discovering or verifying information for use in a book, programme, etc.

Teacher as a researcher

The concept of teacher-as-researcher is included in recent literature on educational reform, which encourages teachers to be collaborators in revising curriculum, improving their work environment, professionalizing teaching, and developing policy. One of the many benefits of teachers researching their own practices is that the teacher has a sense of ownership and control of the research because what is being researched occurs in that teacher's own classroom.

Research in education

Teachers are researchers. They collect enormous amounts of data each day, and they rapidly evaluate and make decisions based on this data. Some of this work is numerical, but much is qualitative. Teachers may be second only to doctors in doing this. What teachers are not good at is doing anything formal with this data. Like a cup sitting under a running tap, more and more information is constantly flowing in, too much to hold. All this data could, however, be used as evidence to inform practice.



A teacher who is a self-directed learner and researcher themselves makes a great role-model to children and colleagues as well as tending to be a better teacher with their ever-evolving understanding of pedagogy.

A teacher can play a valuable role in conducting research, especially in the field of education. As a researcher, a teacher can:

1. **Identify research questions:** Through their experience in the classroom, teachers can identify areas of interest or concerns that they feel could benefit from further investigation. They can formulate research questions that are relevant to their teaching practice and the needs of their students.
2. **Design research studies:** Teachers can design research studies to explore their research questions. They can determine the appropriate research methodology, develop data collection tools, and plan the implementation of the study. This may involve conducting surveys, interviews, observations, or experiments.
3. **Collect and analyze data:** Teachers can gather data using various methods and tools. They can collect information on student performance, behavior, or attitudes, or they

Research in education

can gather data on instructional strategies or interventions. Teachers can also analyze the collected data using statistical techniques to draw meaningful conclusions.

4. **Implement interventions:** Based on their research findings, teachers can develop and implement interventions or strategies to address issues identified through their research. By testing these interventions in their own classrooms, teachers can evaluate their effectiveness and provide evidence-based recommendations for improvement.
5. **Disseminate findings:** Teachers can share their research findings with the educational community through academic journals, conferences, or presentations. By disseminating their research, teachers can contribute to the knowledge base in the field of education and inform other educators about effective strategies and interventions.

Overall, teachers as researchers can bring a unique perspective to educational research by being intimately involved in the day-to-day realities of teaching. Their research can have a direct impact on their own practice as well as on the wider educational community.

2.3 Research ethics

Research ethics

Research ethics refers to the ethical principles and guidelines that govern the conduct of research involving human subjects or animal subjects. It involves ensuring that research is conducted in an ethical manner, with respect for the rights and welfare of the subjects involved.

1. **Informed Consent:** Discuss each and every thing with participants!
Researchers must obtain informed consent from participants, meaning that participants need to be fully aware of the purpose, procedures, risks, and benefits of the research and voluntarily agree to participate.
2. **Privacy and Confidentiality:**
Researchers must ensure the privacy and confidentiality of participants' personal information. They should take measures to protect participants' identities and ensure that data are stored securely.
3. **Beneficence:**

Research in education

Researchers should prioritize the well-being and welfare of research participants. They should minimize potential risks and maximize potential benefits associated with the research.

✓ 4. Non-maleficence:

Researchers should not cause harm to participants. They should take precautions to minimize any potential physical, psychological, or emotional harm or discomfort.

✓ 5. Data Manipulation and Fabrication:

Researchers should not manipulate or fabricate data. They should report their findings truthfully and accurately.

✓ 6. Conflict of Interest:

Researchers should disclose any potential conflicts of interest that could compromise the objectivity and integrity of their research.

✓ 7. Data Manipulation and Fabrication:

Researchers should not manipulate or fabricate data. They should report their findings truthfully and accurately.

✓ 8. Integrity and Objectivity: Researchers must ensure the accuracy and honesty of their research findings, avoid bias or conflict of interest, and report their results without distortion or misrepresentation.

✓ 9. Respect for Ethical Guidelines: Researchers should adhere to national and international ethical guidelines, such as the Declaration of Helsinki for medical research, and obtain ethical approval from Institutional Review Boards or Ethics Review Committees.

✓ 10. Animal Welfare: Research involving animal subjects should follow ethical guidelines to minimize pain, suffering, and distress. Researchers should use alternatives to animal testing whenever possible and ensure that animals are treated humanely.

✓ 11. Responsible Conduct: Researchers should conduct their research responsibly, including proper data management, publication, and sharing of research findings, as well as addressing any ethical concerns raised during the research process.

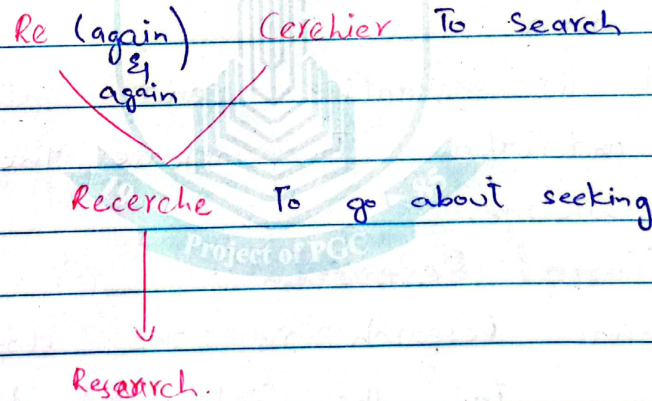
Failure to adhere to research ethics can have serious consequences, including harm to participants, damage to research integrity, legal and professional ramifications, and loss of public trust in the scientific community. Therefore, it is essential for researchers to prioritize ethical considerations throughout the research process.

Chp #1

Educational Research

Examiner's
Remarks

Derived from. The word research is derived from the Middle French word *Recherche* which means "To go about ^{seeking}". The term itself being derived from Old French Term *Reachier* a compound Term (which combines Two or more than Two words) from *Re+Cerchier* which means *Search*.



The earliest use of this Term in 1577.

Definitions of Research

- Research is the simplest Terms in searching for Truth & knowledge.
- Research is defined as the creation of new knowledge in already existing knowledge.
- Research is the systematic investigation and the study of material and sources in order to establish facts and reach new conclusions.
- Research is creative and systematic work undertaken to increase the stock of knowledge.

Educational Research

- Educational Research is the study and investigation in the field of Education.
C.V. GOOD.
- The final purpose of educational research is to ascertain principles

and develop procedures for use in field of education. Munroe.

→ Educational Research is a process in which educational problems are solved with the help of investigation.

→ Educational Research refers to systematic collection and analysis of a data related to the field of Education.

Scope of study →

Scope of study means in which the research area will be explored in the work.

Scope is the extent of the area or subject of matter.

• Sample •

A sample is defined as a smaller set of data that is chosen or selected from a larger population.

• Population •

A population is the entire group that you want to draw conclusions about. A population refers to any collection specified group of human beings. Population commonly contain two many individuals to study conveniently.

• Sampling •

Sampling means selecting the group that you will actually collect data from in your research.

Sampling is the process of studying the population by gathering information and

analyzing that data.

Inferential Analysis

- Summarize your data by using graphs.
- Draw a conclusion about population through a sample.
- All conclusion about population through a sample.
- Use sample data to know about the population.
- Using data collected from a small group to draw conclusions about a larger group.

Difference b/w Descriptive

Examiner's
Remarks

Analysis

- 1- **Mean** $\longrightarrow$ Average Number of Data
- 2- **Mode** $\longrightarrow$ Most Repeated Number of Data.
- 3- **Median** $\longrightarrow$ Data divide into 2 parts.
- 4- **Quartiles** $\longrightarrow$ Data divide into 4 parts.
- 5- **Deciles** $\longrightarrow$ Data divide into 10 parts.
- 6- **Percentiles** $\longrightarrow$ Data divide into 100 parts.
- 7- **Range** $\longrightarrow$ The difference b/w highest & lowest value of Data.
- 8- **Quartile Deviation** $\longrightarrow$ The ^{half of} difference between upper and lower quartile.
- 9- **Standard Deviation** $\longrightarrow$ Describe how (spread widely) dispersed a ^{set of} data is.

Sociology Chp#1:- Research Methodology

1.3 Relationship B/w Theory & Research.

Theory

Research

- * A theory is a way to explain different aspects of social trends, of social structures, dynamics and principles that social interactions exists between individuals and within societies.
- * A well-structure set of ideas that explain and analyses society, social behaviour by following a systematic and social phenomena plan.
- * It is not a random ^{just} Hunch & guess; Sociological Theories based on Research, observation & Data.
- * It is an inter-related sets of law-like propositions subject to their empirical verification.
- * The goal of a social theory is to formulate universal propositions reflecting the social reality but subject to their empirical verification by an average researcher.
- * Social research is the study of social trends, dynamics and principles that exists between individuals and within societies.
- * Social Research is a method used by social scientists following a systematic and social phenomena plan.
- * Social Research Methodologies are considered or classified as quantitative & qualitative.
- * Social Research is a systematic and scientific way of studying Human society and social interaction.
- * Social Research is the systematic method of discovering new facts, or verifying old facts.
- * The primary goal of research is to understand social life and thereby gain a greater measure of control over social Behaviour.

Relationship

Theory and research in a social science have a reciprocal relationship meaning they influence each other in a "Back-forth way". There relation is "dynamic" and "interrelated". "Cyclical" and "Secondary Process".

Theory Guides Research.

- * Guides framework
- * Theoretical Framework
- * Research Questions.

⇒ state of what causes what, in what circumstances and why.

Research informs Theory.

- * Test and Refines Theories.
- * Discover New Theories

⇒ Research is all about addressing an issue or asking and answering a question or solving a Problem.

Inductive Way (Qualitative)
Research → Theory

Deductive Way (Quantitative)
Theory → Research

1. * Theory is like a map, guiding researchers on their investigative Journey. Research help in refining and updating the map, ensuring it accurately reflects the explored Territory.